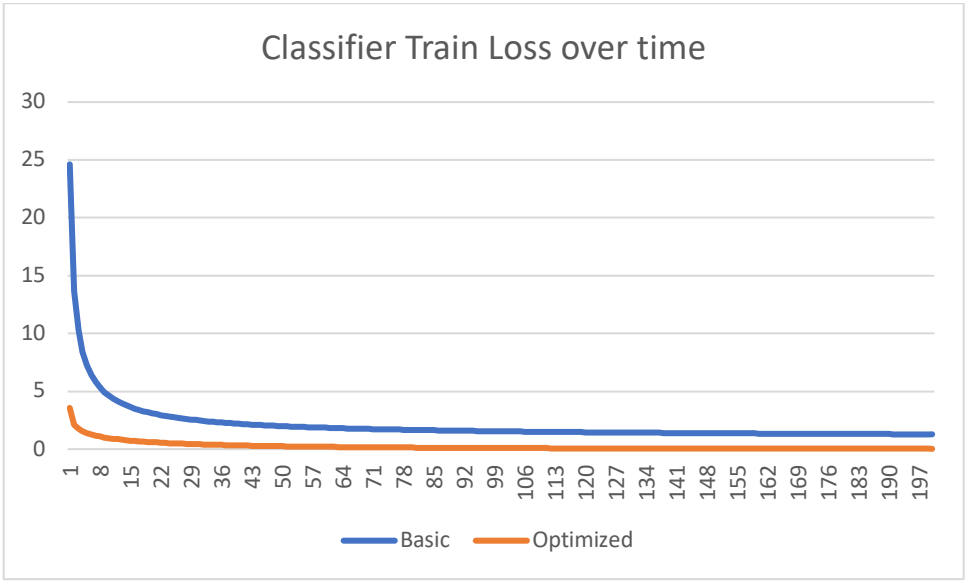
Neural Network Results Plotted:



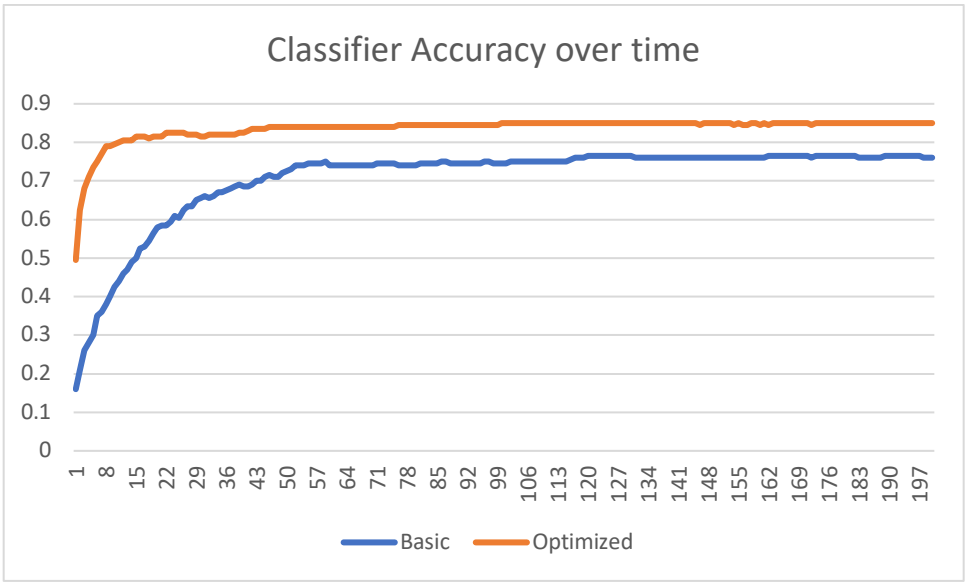


Image classification using least squares method and neural network - technical report

Maxine Perroni-Scharf, 03/09/2020

This report documents the methodology, code implementation and results for two approaches to image classification: a) a least squares solver and b) a neural network. We will be training and testing our classification algorithms using the MNIST_Sub dataset of handwritten digits.

Final Results

Least Squares Function Approximation: MSE = 0

Least Squares Image Classification: Accuracy = 49.5%

Neural Network: Accuracy = 85.0%

Time taken for single SolveLeastSquares (i.e. calculate single A^TA and A^Tb and solve): Time = 7.5s

Least Squares Solver

As discussed in class, I followed the standard methodology for implementing a least squares data training algorithm.

Function Approximator

For my function approximator, I wrote a method that takes in a max degree and creates a matrix A of the where each row is of the form (1, x, x^2, ... x^maxdegree).

```
Matrix CreatePolynomialA(const Matrix& xData, int degree) {
Matrix A(xData.m, degree + 1);
for (int i = 0; i < xData.m; i ++) {
for (int j = 0; j < degree + 1; j ++) {
A(i, j) = pow(xData(i, 0), j);
}
}
return A;
}
```

For example, here is the output of CreatePolynomialA([1, 2, 3, 4, 5], 3):

```
Checking Data Matrix Generation:

1, 1, 1, 1,
1, 2, 4, 8,
1, 3, 9, 27,
1, 4, 16, 64,
1, 5, 25, 125,
```

This allowed me to test various different functions to approximate my data easily (rather than just hardcoding in f1, f2 etc).

In order to test my approximator, I generated a training and testing set of data with random x values and y values which correspond to an exact function. This meant that I knew if my approximator found the exact theta values that matched the coefficients of the function I chose to calculate the y values, then my approximator was working correctly. I got an **MSE value of 0** using this method (which is expected, as my training data and test data all lie exactly on a real polynomial line).

Image classification

In order to use least squares for image classification, I wrote methods to read in the data set values from the MNIST_Sub directory. I did this by reading line by line and adding in the 784 pixel values and one bias term (1) to each row of my matrix A. I also added in a random noise value to each entry of A to ensure the matrix A was not singular when solving A^TAx = A^Tb.

```
 // Read in training data from directory
Matrix GenerateData(const string& data_dict, int num_data_rows) {
    Matrix train_data(num_data_rows, 785);
    int tempint;
    ifstream file(data_dict+"/train_data.txt");
    if (!file.is_open()) std::cout<<"file does not exist"<<std::endl;

// For every row in file, get value
for(int i=0;i<num_data_rows;i++) {
    for (int j = 0; j < 784; j ++) {
        if(file >> tempint) {
            train_data(i, j)=(double)tempint/255.0;
        }
    }
}
file.close();

// Add random value to each to prevent singularity
for (int i = 0; i < train_data.m; i ++) {
    for (int j = 0; j < train_data.n; j ++) {
        train_data(i, j) += rand()%10 * 0.011;
    }
    train_data(i, 784) = 1;
}
return train_data;
}
```

I also wrote functions for one hot encoding the labels of the training data and for getting the real values back from the encoded values, and then trained each column of theta one at a time, so that each column of theta can determine whether or not an image is a given digit or not.

```
// Convert one_hot encoded back to real values
Matrix getRealVals(Matrix& encoded) {
    Matrix real(encoded.m, 1);
    for (int i = 0; i < real.m; i ++) {
        double max = -100;
        int index = 0;

        // value = argmax(matrix row)
        for (int j = 0; j < encoded.n; j ++) {
            if (encoded(i, j) > max) {
                index = j;
                max = encoded(i, j);
            }
        }
        real(i, 0) = index;
    }
    return real;
}
```

I then tested my approximator using the MNIST_Sub data set of images as discussed in class using a driver method. I wrote an accuracy function to evaluate this performance.

```
// Check how many are correct
double GetAccuracy(Matrix& A, Matrix& B) {
    double total = 0;
    for (int i = 0; i < A.m; i ++) {
        if (A(i, 0) == B(i, 0)) {
            total += 1;
        }
    }
    return total/A.m;
}
```

Using this dataset, I was able to predict the labels of the test images with a**49.5%** accuracy, which is significantly better than randomly guessing the images, however far below the desired accuracy for a successful classification algorithm. It took approximately 1 minute 30 seconds to produce this result.

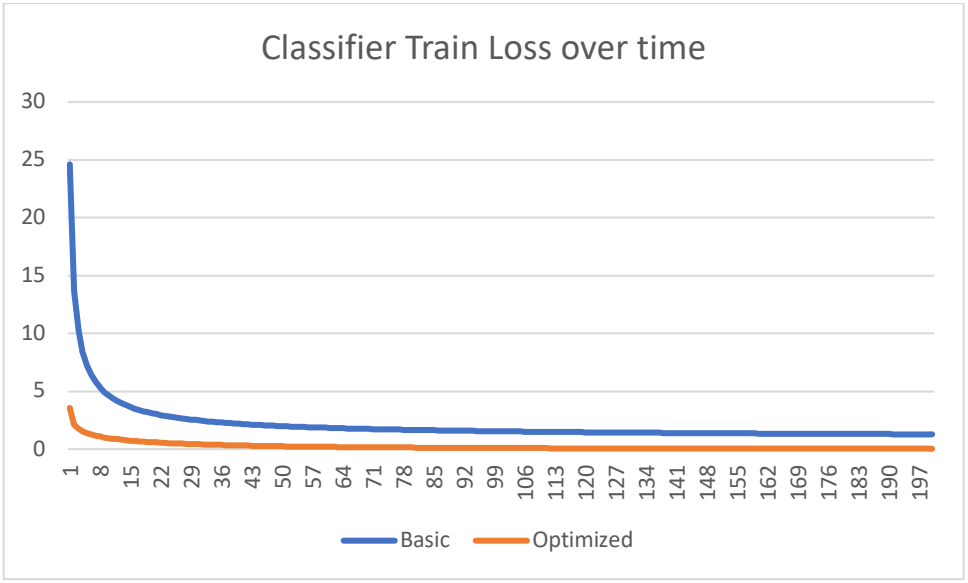
Challenges

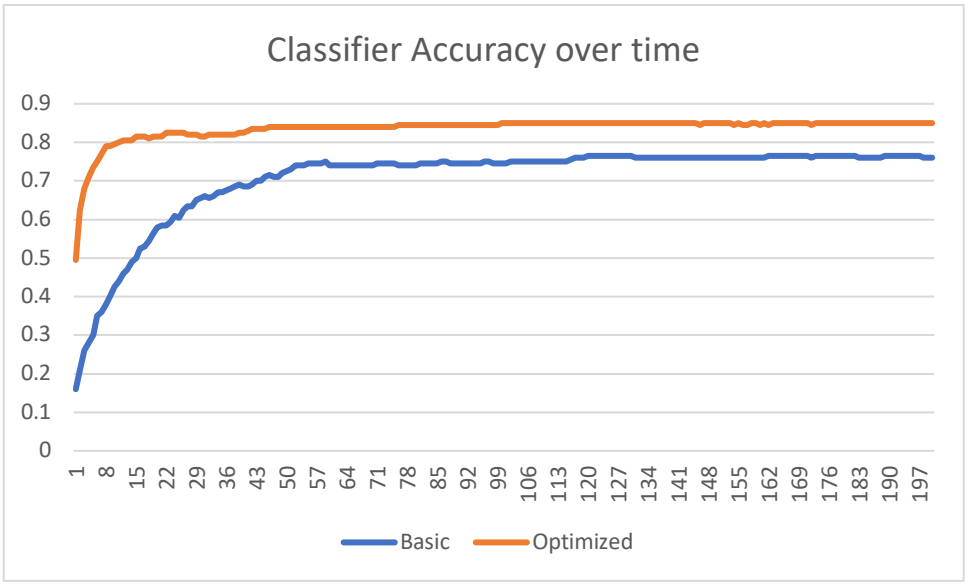
One challenge I faced during this implementation was reading in and one_hot encoding the labels in the most efficient way possible.

The other challenge I faced was that my approximator took 5+ minutes to run, so in order to fix this I went back through my matrix library to optimize my code there.

Neural Network

Neural Network Results Plotted:





Using the starter code for the neural network, I populated the forward and backwards methods for the nonlinear activation class and the loss class.

Challenges

The challenges I faced during implementing the basic neural network were mainly due to small bugs I found in my matrix library method. I also struggled a little figuring out the order of linear and non linear layers in forwards and backwards propagation, and had many bugs because my order was the wrong way around.

Using the basic neural network, I was able to get an accuracy of**75.5%** in my image classification, which is significantly better than the results from the least squares solver.

Extra Features

New Network Architecture

I implemented three extra features in the neural network code. Firstly, in the forward and backwards passes, I tried using a different architecture where there is one linear layer, and then two activation layers, and then one linear layer etc. I implemented this by passing the data through each activation layer twice instead of once in forward and backward propagation.

```
for (int i = linear_layers.size() - 2; i > -1; i --) {

# Use each activation layer twice
current = this->activation_layers[i].Backward(current);
current = this->activation_layers[i].Backward(current);

current =this->linear_layers[i].Backward(current);

}
```

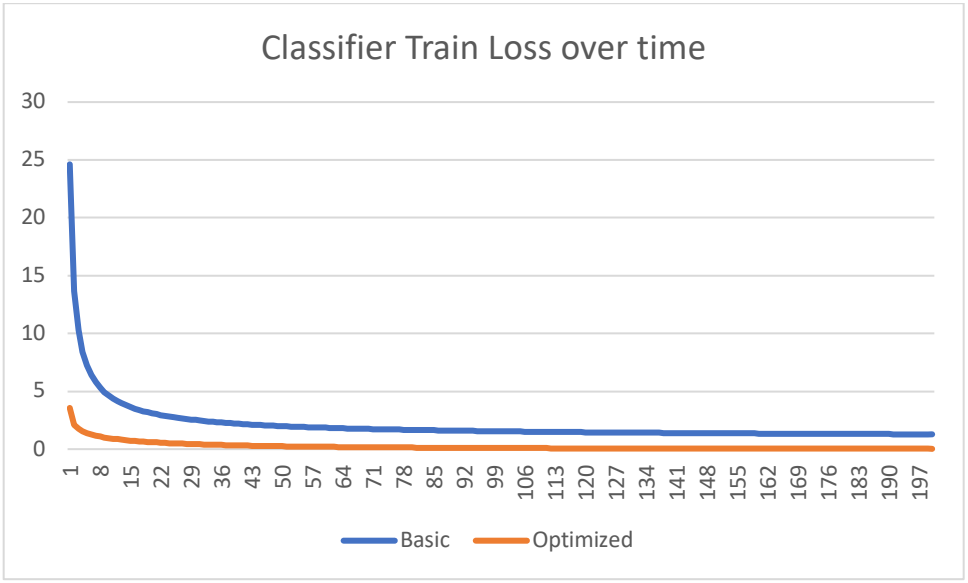
This was able to increase my accuracy by 0.5% when combined with my new activation function. This did not have an effect on my code until after I implemented the new activation function, as doing standard ReLU once is the same as doing it twice.

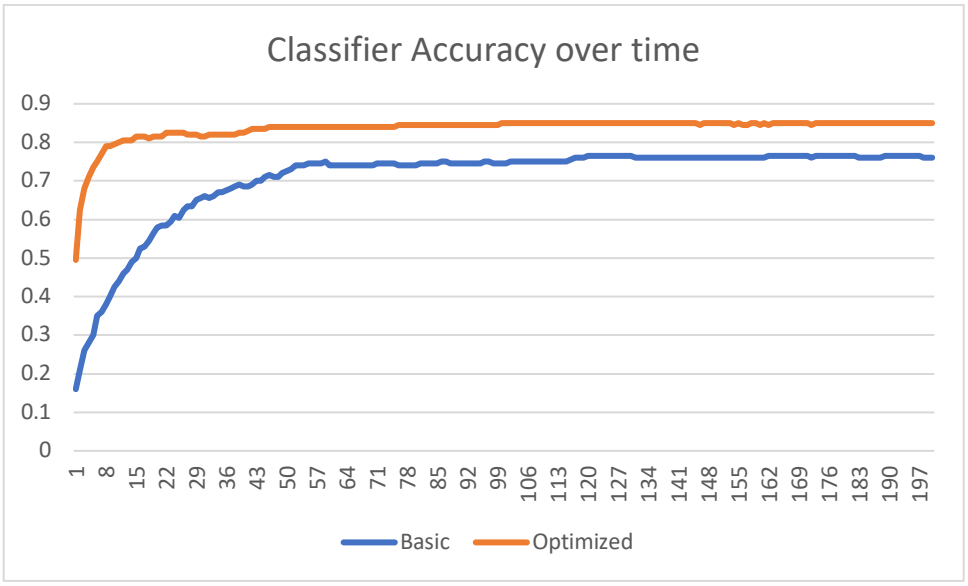
New Activation Function

I tried several different activation functions, and the most successful one was very similar to ReLU but instead of using max(0, x) I used max (0.1*x, x). Through this adjustment and by changing the learning rate to 1e-2, my accuracy was increased further to **85%** which is a huge improvement for a very small change.

The code for the Forward method of this new function is as follows:

Neural Network Results Plotted:





```
 Matrix relu = input;

for (int i = 0; i < relu.m; i ++) {
    for (int j = 0; j < relu.n; j ++) {
        if (input(i,j) > 0) {
            relu(i, j) = input(i, j);
        }
        else {
            relu(i, j) = input(i, j) * 0.1;
        }
    }
}
this->stored_input = relu;
return relu;
```

The new network architecture and new activation function are implemented in MyNeuralNetworkDessert.h and can be tested with HW_ID = 8.

Bias

I added in a column of 1s as a bias term to the matrix A (so instead of being 1000x784, A is now 1000x785). However, the accuracy after adding in this bias term was actually worse than the original implementation, so I decided not to include this in my final code (it is implemented in MyNeuralNetworkBias.h and can be tested with HW_ID = 9).

Testing

To test my neural network please use HW_ID = 8 in the main function, and to test least squares use HW_ID = 11.

Outputs using optimized neural network code

```
 Epoch: 1/200 | Train loss: 9.88339 | Test loss: 39.7705
Epoch: 2/200 | Train loss: 6.3277 | Test loss: 39.7653
Epoch: 3/200 | Train loss: 6.32767 | Test loss: 39.765
Epoch: 4/200 | Train loss: 6.32767 | Test loss: 39.7647
Epoch: 5/200 | Train loss: 6.32768 | Test loss: 39.7644
Epoch: 6/200 | Train loss: 6.32768 | Test loss: 39.7641
Epoch: 7/200 | Train loss: 6.32768 | Test loss: 39.7638
Epoch: 8/200 | Train loss: 6.32768 | Test loss: 39.7635
Epoch: 9/200 | Train loss: 6.32762 | Test loss: 39.7668
Epoch: 10/200 | Train loss: 6.32767 | Test loss: 39.7628
Epoch: 11/200 | Train loss: 6.32768 | Test loss: 39.7626
Epoch: 12/200 | Train loss: 6.32762 | Test loss: 39.7659
Epoch: 13/200 | Train loss: 6.32768 | Test loss: 39.7619
Epoch: 14/200 | Train loss: 6.32769 | Test loss: 39.7617
Epoch: 15/200 | Train loss: 6.32763 | Test loss: 39.765
```

```
Epoch: 16/200 | Train loss: 6.32768 | Test loss: 39.761
Epoch: 17/200 | Train loss: 6.32769 | Test loss: 39.7608
Epoch: 18/200 | Train loss: 6.32763 | Test loss: 39.7642
Epoch: 19/200 | Train loss: 6.32768 | Test loss: 39.7602
Epoch: 20/200 | Train loss: 6.3277 | Test loss: 39.76
Epoch: 21/200 | Train loss: 6.32764 | Test loss: 39.7633
Epoch: 22/200 | Train loss: 6.32769 | Test loss: 39.7593
Epoch: 23/200 | Train loss: 6.3277 | Test loss: 39.7591
Epoch: 24/200 | Train loss: 6.3277 | Test loss: 39.7588
Epoch: 25/200 | Train loss: 6.32764 | Test loss: 39.7621
Epoch: 26/200 | Train loss: 6.3277 | Test loss: 39.7581
Epoch: 27/200 | Train loss: 6.32771 | Test loss: 39.7579
Epoch: 28/200 | Train loss: 6.32764 | Test loss: 39.7612
Epoch: 29/200 | Train loss: 6.3277 | Test loss: 39.7572
Epoch: 30/200 | Train loss: 6.32771 | Test loss: 39.757
Epoch: 31/200 | Train loss: 6.32765 | Test loss: 39.7604
Epoch: 32/200 | Train loss: 6.32771 | Test loss: 39.7563
Epoch: 33/200 | Train loss: 6.32772 | Test loss: 39.7561
Epoch: 34/200 | Train loss: 6.32765 | Test loss: 39.7595
Epoch: 35/200 | Train loss: 6.32771 | Test loss: 39.7554
Epoch: 36/200 | Train loss: 6.32772 | Test loss: 39.7552
Epoch: 37/200 | Train loss: 6.32766 | Test loss: 39.7586
Epoch: 38/200 | Train loss: 6.32771 | Test loss: 39.7546
Epoch: 39/200 | Train loss: 6.32773 | Test loss: 39.7544
Epoch: 40/200 | Train loss: 6.32766 | Test loss: 39.7577
Epoch: 41/200 | Train loss: 6.32772 | Test loss: 39.7537
Epoch: 42/200 | Train loss: 6.32773 | Test loss: 39.7535
Epoch: 43/200 | Train loss: 6.32767 | Test loss: 39.7569
Epoch: 44/200 | Train loss: 6.32772 | Test loss: 39.7528
Epoch: 45/200 | Train loss: 6.32774 | Test loss: 39.7526
Epoch: 46/200 | Train loss: 6.32774 | Test loss: 39.7523
Epoch: 47/200 | Train loss: 6.32767 | Test loss: 39.7557
Epoch: 48/200 | Train loss: 6.32773 | Test loss: 39.7516
Epoch: 49/200 | Train loss: 6.32774 | Test loss: 39.7514
Epoch: 50/200 | Train loss: 6.32768 | Test loss: 39.7548
Epoch: 51/200 | Train loss: 6.32774 | Test loss: 39.7507
Epoch: 52/200 | Train loss: 6.32775 | Test loss: 39.7505
Epoch: 53/200 | Train loss: 6.32768 | Test loss: 39.7539
Epoch: 54/200 | Train loss: 6.32774 | Test loss: 39.7498
Epoch: 55/200 | Train loss: 6.32775 | Test loss: 39.7497
Epoch: 56/200 | Train loss: 6.32769 | Test loss: 39.7531
Epoch: 57/200 | Train loss: 6.32773 | Test loss: 39.7493
Epoch: 58/200 | Train loss: 6.32776 | Test loss: 39.7488
Epoch: 59/200 | Train loss: 6.32769 | Test loss: 39.7522
Epoch: 60/200 | Train loss: 6.32775 | Test loss: 39.7481
Epoch: 61/200 | Train loss: 6.32776 | Test loss: 39.7479
Epoch: 62/200 | Train loss: 6.3277 | Test loss: 39.7513
Epoch: 63/200 | Train loss: 6.32776 | Test loss: 39.7472
Epoch: 64/200 | Train loss: 6.32777 | Test loss: 39.747
```

```
Epoch: 64/200 | Train loss: 6.32777 | Test loss: 39.747
Epoch: 65/200 | Train loss: 6.32771 | Test loss: 39.7504
Epoch: 66/200 | Train loss: 6.32776 | Test loss: 39.7463
Epoch: 67/200 | Train loss: 6.32777 | Test loss: 39.7461
Epoch: 68/200 | Train loss: 6.32777 | Test loss: 39.7458
Epoch: 69/200 | Train loss: 6.32771 | Test loss: 39.7493
Epoch: 70/200 | Train loss: 6.32777 | Test loss: 39.7451
Epoch: 71/200 | Train loss: 6.32778 | Test loss: 39.745
Epoch: 72/200 | Train loss: 6.32771 | Test loss: 39.7484
Epoch: 73/200 | Train loss: 6.32778 | Test loss: 39.7442
Epoch: 74/200 | Train loss: 6.32778 | Test loss: 39.7441
Epoch: 75/200 | Train loss: 6.32772 | Test loss: 39.7475
Epoch: 76/200 | Train loss: 6.32778 | Test loss: 39.7434
Epoch: 77/200 | Train loss: 6.32779 | Test loss: 39.7432
Epoch: 78/200 | Train loss: 6.32773 | Test loss: 39.7467
Epoch: 79/200 | Train loss: 6.32778 | Test loss: 39.7425
Epoch: 80/200 | Train loss: 6.32779 | Test loss: 39.7423
Epoch: 81/200 | Train loss: 6.32773 | Test loss: 39.7458
Epoch: 82/200 | Train loss: 6.32779 | Test loss: 39.7416
Epoch: 83/200 | Train loss: 6.3278 | Test loss: 39.7414
Epoch: 84/200 | Train loss: 6.32774 | Test loss: 39.7449
Epoch: 85/200 | Train loss: 6.32779 | Test loss: 39.7407
Epoch: 86/200 | Train loss: 6.32781 | Test loss: 39.7406
Epoch: 87/200 | Train loss: 6.32781 | Test loss: 39.7403
Epoch: 88/200 | Train loss: 6.32774 | Test loss: 39.7438
Epoch: 89/200 | Train loss: 6.3278 | Test loss: 39.7396
Epoch: 90/200 | Train loss: 6.32781 | Test loss: 39.7394
Epoch: 91/200 | Train loss: 6.32775 | Test loss: 39.7429
Epoch: 92/200 | Train loss: 6.32781 | Test loss: 39.7387
Epoch: 93/200 | Train loss: 6.32782 | Test loss: 39.7385
Epoch: 94/200 | Train loss: 6.32775 | Test loss: 39.742
Epoch: 95/200 | Train loss: 6.32781 | Test loss: 39.7378
Epoch: 96/200 | Train loss: 6.32782 | Test loss: 39.7376
Epoch: 97/200 | Train loss: 6.32776 | Test loss: 39.7412
Epoch: 98/200 | Train loss: 6.32782 | Test loss: 39.7369
Epoch: 99/200 | Train loss: 6.32783 | Test loss: 39.7368
Epoch: 100/200 | Train loss: 6.32776 | Test loss: 39.7403
Epoch: 101/200 | Train loss: 6.32782 | Test loss: 39.7361
Epoch: 102/200 | Train loss: 6.32783 | Test loss: 39.7359
Epoch: 103/200 | Train loss: 6.32777 | Test loss: 39.7394
Epoch: 104/200 | Train loss: 6.32783 | Test loss: 39.7352
Epoch: 105/200 | Train loss: 6.32784 | Test loss: 39.735
Epoch: 106/200 | Train loss: 6.32784 | Test loss: 39.7347
Epoch: 107/200 | Train loss: 6.32777 | Test loss: 39.7383
Epoch: 108/200 | Train loss: 6.32784 | Test loss: 39.734
Epoch: 109/200 | Train loss: 6.32785 | Test loss: 39.7339
Epoch: 110/200 | Train loss: 6.32778 | Test loss: 39.7374
Epoch: 111/200 | Train loss: 6.32784 | Test loss: 39.7331
Epoch: 112/200 | Train loss: 6.32785 | Test loss: 39.733
```

```
Epoch: 113/200 | Train loss: 6.32779 | Test loss: 39.7365
Epoch: 114/200 | Train loss: 6.32785 | Test loss: 39.7323
Epoch: 115/200 | Train loss: 6.32786 | Test loss: 39.7321
Epoch: 116/200 | Train loss: 6.32779 | Test loss: 39.7357
Epoch: 117/200 | Train loss: 6.32785 | Test loss: 39.7314
Epoch: 118/200 | Train loss: 6.32786 | Test loss: 39.7312
Epoch: 119/200 | Train loss: 6.3278 | Test loss: 39.7348
Epoch: 120/200 | Train loss: 6.32786 | Test loss: 39.7305
Epoch: 121/200 | Train loss: 6.32787 | Test loss: 39.7304
Epoch: 122/200 | Train loss: 6.32787 | Test loss: 39.7301
Epoch: 123/200 | Train loss: 6.3278 | Test loss: 39.7337
Epoch: 124/200 | Train loss: 6.32787 | Test loss: 39.7294
Epoch: 125/200 | Train loss: 6.32788 | Test loss: 39.7292
Epoch: 126/200 | Train loss: 6.32781 | Test loss: 39.7328
Epoch: 127/200 | Train loss: 6.32787 | Test loss: 39.7285
Epoch: 128/200 | Train loss: 6.32788 | Test loss: 39.7283
Epoch: 129/200 | Train loss: 6.32781 | Test loss: 39.7319
Epoch: 130/200 | Train loss: 6.32788 | Test loss: 39.7276
Epoch: 131/200 | Train loss: 6.32789 | Test loss: 39.7275
Epoch: 132/200 | Train loss: 6.32782 | Test loss: 39.7311
Epoch: 133/200 | Train loss: 6.32788 | Test loss: 39.7268
Epoch: 134/200 | Train loss: 6.32789 | Test loss: 39.7266
Epoch: 135/200 | Train loss: 6.32783 | Test loss: 39.7302
Epoch: 136/200 | Train loss: 6.32789 | Test loss: 39.7259
Epoch: 137/200 | Train loss: 6.3279 | Test loss: 39.7257
Epoch: 138/200 | Train loss: 6.32789 | Test loss: 39.7257
Epoch: 139/200 | Train loss: 6.32783 | Test loss: 39.7291
Epoch: 140/200 | Train loss: 6.3279 | Test loss: 39.7247
Epoch: 141/200 | Train loss: 6.32791 | Test loss: 39.7246
Epoch: 142/200 | Train loss: 6.32784 | Test loss: 39.7282
Epoch: 143/200 | Train loss: 6.3279 | Test loss: 39.7239
Epoch: 144/200 | Train loss: 6.32791 | Test loss: 39.7237
Epoch: 145/200 | Train loss: 6.32784 | Test loss: 39.7273
Epoch: 146/200 | Train loss: 6.32791 | Test loss: 39.723
Epoch: 147/200 | Train loss: 6.32792 | Test loss: 39.7229
Epoch: 148/200 | Train loss: 6.32785 | Test loss: 39.7265
Epoch: 149/200 | Train loss: 6.32791 | Test loss: 39.7221
Epoch: 150/200 | Train loss: 6.32792 | Test loss: 39.722
Epoch: 151/200 | Train loss: 6.32786 | Test loss: 39.7256
Epoch: 152/200 | Train loss: 6.32792 | Test loss: 39.7213
Epoch: 153/200 | Train loss: 6.32793 | Test loss: 39.7211
Epoch: 154/200 | Train loss: 6.32793 | Test loss: 39.7208
Epoch: 155/200 | Train loss: 6.32786 | Test loss: 39.7245
Epoch: 156/200 | Train loss: 6.32793 | Test loss: 39.7201
Epoch: 157/200 | Train loss: 6.32794 | Test loss: 39.72
Epoch: 158/200 | Train loss: 6.32787 | Test loss: 39.7236
Epoch: 159/200 | Train loss: 6.32794 | Test loss: 39.7193
Epoch: 160/200 | Train loss: 6.32794 | Test loss: 39.7191
Epoch: 161/200 | Train loss: 6.32787 | Test loss: 39.7228
```

```
Epoch: 162/200 | Train loss: 6.32794 | Test loss: 39.7184
Epoch: 163/200 | Train loss: 6.32795 | Test loss: 39.7183
Epoch: 164/200 | Train loss: 6.32788 | Test loss: 39.7219
Epoch: 165/200 | Train loss: 6.32795 | Test loss: 39.7175
Epoch: 166/200 | Train loss: 6.32795 | Test loss: 39.7174
Epoch: 167/200 | Train loss: 6.32789 | Test loss: 39.7211
Epoch: 168/200 | Train loss: 6.32795 | Test loss: 39.7167
Epoch: 169/200 | Train loss: 6.32796 | Test loss: 39.7165
Epoch: 170/200 | Train loss: 6.32796 | Test loss: 39.7162
Epoch: 171/200 | Train loss: 6.32789 | Test loss: 39.7199
Epoch: 172/200 | Train loss: 6.32796 | Test loss: 39.7155
Epoch: 173/200 | Train loss: 6.32797 | Test loss: 39.7154
Epoch: 174/200 | Train loss: 6.3279 | Test loss: 39.7191
Epoch: 175/200 | Train loss: 6.32797 | Test loss: 39.7147
Epoch: 176/200 | Train loss: 6.32797 | Test loss: 39.7145
Epoch: 177/200 | Train loss: 6.3279 | Test loss: 39.7182
Epoch: 178/200 | Train loss: 6.32797 | Test loss: 39.7138
Epoch: 179/200 | Train loss: 6.32798 | Test loss: 39.7137
Epoch: 180/200 | Train loss: 6.32791 | Test loss: 39.7174
Epoch: 181/200 | Train loss: 6.32798 | Test loss: 39.713
Epoch: 182/200 | Train loss: 6.32799 | Test loss: 39.7128
Epoch: 183/200 | Train loss: 6.32792 | Test loss: 39.7165
Epoch: 184/200 | Train loss: 6.32798 | Test loss: 39.7121
Epoch: 185/200 | Train loss: 6.32799 | Test loss: 39.712
Epoch: 186/200 | Train loss: 6.328 | Test loss: 39.7117
Epoch: 187/200 | Train loss: 6.32792 | Test loss: 39.7154
Epoch: 188/200 | Train loss: 6.32799 | Test loss: 39.711
Epoch: 189/200 | Train loss: 6.328 | Test loss: 39.7108
Epoch: 190/200 | Train loss: 6.32793 | Test loss: 39.7146
Epoch: 191/200 | Train loss: 6.328 | Test loss: 39.7101
Epoch: 192/200 | Train loss: 6.32801 | Test loss: 39.71
Epoch: 193/200 | Train loss: 6.32794 | Test loss: 39.7137
Epoch: 194/200 | Train loss: 6.328 | Test loss: 39.7093
Epoch: 195/200 | Train loss: 6.32801 | Test loss: 39.7091
Epoch: 196/200 | Train loss: 6.32794 | Test loss: 39.7129
Epoch: 197/200 | Train loss: 6.32801 | Test loss: 39.7084
Epoch: 198/200 | Train loss: 6.32802 | Test loss: 39.7083
Epoch: 199/200 | Train loss: 6.32802 | Test loss: 39.708
Epoch: 200/200 | Train loss: 6.32795 | Test loss: 39.7117
Epoch: 1/200 | Train loss: 3.56129 | Test Accuracy: 0.495
Epoch: 2/200 | Train loss: 2.11836 | Test Accuracy: 0.625
Epoch: 3/200 | Train loss: 1.73989 | Test Accuracy: 0.68
Epoch: 4/200 | Train loss: 1.52135 | Test Accuracy: 0.71
Epoch: 5/200 | Train loss: 1.36885 | Test Accuracy: 0.735
Epoch: 6/200 | Train loss: 1.2512 | Test Accuracy: 0.75
Epoch: 7/200 | Train loss: 1.15719 | Test Accuracy: 0.77
Epoch: 8/200 | Train loss: 1.07862 | Test Accuracy: 0.79
Epoch: 9/200 | Train loss: 1.01156 | Test Accuracy: 0.79
Epoch: 10/200 | Train loss: 0.952648 | Test Accuracy: 0.795
```

```
Epoch: 10/200 | Train loss: 0.952648 | Test Accuracy: 0.795
Epoch: 11/200 | Train loss: 0.900417 | Test Accuracy: 0.8
Epoch: 12/200 | Train loss: 0.85354 | Test Accuracy: 0.805
Epoch: 13/200 | Train loss: 0.811439 | Test Accuracy: 0.805
Epoch: 14/200 | Train loss: 0.772962 | Test Accuracy: 0.805
Epoch: 15/200 | Train loss: 0.737949 | Test Accuracy: 0.815
Epoch: 16/200 | Train loss: 0.705818 | Test Accuracy: 0.815
Epoch: 17/200 | Train loss: 0.675922 | Test Accuracy: 0.815
Epoch: 18/200 | Train loss: 0.648264 | Test Accuracy: 0.81
Epoch: 19/200 | Train loss: 0.622615 | Test Accuracy: 0.815
Epoch: 20/200 | Train loss: 0.59841 | Test Accuracy: 0.815
Epoch: 21/200 | Train loss: 0.575942 | Test Accuracy: 0.815
Epoch: 22/200 | Train loss: 0.554929 | Test Accuracy: 0.825
Epoch: 23/200 | Train loss: 0.535123 | Test Accuracy: 0.825
Epoch: 24/200 | Train loss: 0.516551 | Test Accuracy: 0.825
Epoch: 25/200 | Train loss: 0.498946 | Test Accuracy: 0.825
Epoch: 26/200 | Train loss: 0.482413 | Test Accuracy: 0.825
Epoch: 27/200 | Train loss: 0.466509 | Test Accuracy: 0.82
Epoch: 28/200 | Train loss: 0.451571 | Test Accuracy: 0.82
Epoch: 29/200 | Train loss: 0.437261 | Test Accuracy: 0.82
Epoch: 30/200 | Train loss: 0.423669 | Test Accuracy: 0.815
Epoch: 31/200 | Train loss: 0.410682 | Test Accuracy: 0.815
Epoch: 32/200 | Train loss: 0.398369 | Test Accuracy: 0.82
Epoch: 33/200 | Train loss: 0.38665 | Test Accuracy: 0.82
Epoch: 34/200 | Train loss: 0.375479 | Test Accuracy: 0.82
Epoch: 35/200 | Train loss: 0.364739 | Test Accuracy: 0.82
Epoch: 36/200 | Train loss: 0.354557 | Test Accuracy: 0.82
Epoch: 37/200 | Train loss: 0.34482 | Test Accuracy: 0.82
Epoch: 38/200 | Train loss: 0.335587 | Test Accuracy: 0.82
Epoch: 39/200 | Train loss: 0.326559 | Test Accuracy: 0.825
Epoch: 40/200 | Train loss: 0.317952 | Test Accuracy: 0.825
Epoch: 41/200 | Train loss: 0.309774 | Test Accuracy: 0.83
Epoch: 42/200 | Train loss: 0.30189 | Test Accuracy: 0.835
Epoch: 43/200 | Train loss: 0.294218 | Test Accuracy: 0.835
Epoch: 44/200 | Train loss: 0.286901 | Test Accuracy: 0.835
Epoch: 45/200 | Train loss: 0.279775 | Test Accuracy: 0.835
Epoch: 46/200 | Train loss: 0.272936 | Test Accuracy: 0.84
Epoch: 47/200 | Train loss: 0.266386 | Test Accuracy: 0.84
Epoch: 48/200 | Train loss: 0.259985 | Test Accuracy: 0.84
Epoch: 49/200 | Train loss: 0.253894 | Test Accuracy: 0.84
Epoch: 50/200 | Train loss: 0.24806 | Test Accuracy: 0.84
Epoch: 51/200 | Train loss: 0.242272 | Test Accuracy: 0.84
Epoch: 52/200 | Train loss: 0.23665 | Test Accuracy: 0.84
Epoch: 53/200 | Train loss: 0.231419 | Test Accuracy: 0.84
Epoch: 54/200 | Train loss: 0.226157 | Test Accuracy: 0.84
Epoch: 55/200 | Train loss: 0.221118 | Test Accuracy: 0.84
Epoch: 56/200 | Train loss: 0.216252 | Test Accuracy: 0.84
Epoch: 57/200 | Train loss: 0.211542 | Test Accuracy: 0.84
Epoch: 58/200 | Train loss: 0.206928 | Test Accuracy: 0.84
```

```
Epoch: 59/200 | Train loss: 0.202526 | Test Accuracy: 0.84
Epoch: 60/200 | Train loss: 0.198219 | Test Accuracy: 0.84
Epoch: 61/200 | Train loss: 0.194049 | Test Accuracy: 0.84
Epoch: 62/200 | Train loss: 0.189985 | Test Accuracy: 0.84
Epoch: 63/200 | Train loss: 0.185984 | Test Accuracy: 0.84
Epoch: 64/200 | Train loss: 0.182176 | Test Accuracy: 0.84
Epoch: 65/200 | Train loss: 0.178403 | Test Accuracy: 0.84
Epoch: 66/200 | Train loss: 0.174806 | Test Accuracy: 0.84
Epoch: 67/200 | Train loss: 0.171288 | Test Accuracy: 0.84
Epoch: 68/200 | Train loss: 0.167859 | Test Accuracy: 0.84
Epoch: 69/200 | Train loss: 0.1645 | Test Accuracy: 0.84
Epoch: 70/200 | Train loss: 0.161237 | Test Accuracy: 0.84
Epoch: 71/200 | Train loss: 0.158025 | Test Accuracy: 0.84
Epoch: 72/200 | Train loss: 0.154991 | Test Accuracy: 0.84
Epoch: 73/200 | Train loss: 0.151945 | Test Accuracy: 0.84
Epoch: 74/200 | Train loss: 0.149037 | Test Accuracy: 0.84
Epoch: 75/200 | Train loss: 0.14617 | Test Accuracy: 0.84
Epoch: 76/200 | Train loss: 0.14341 | Test Accuracy: 0.845
Epoch: 77/200 | Train loss: 0.140714 | Test Accuracy: 0.845
Epoch: 78/200 | Train loss: 0.138099 | Test Accuracy: 0.845
Epoch: 79/200 | Train loss: 0.13552 | Test Accuracy: 0.845
Epoch: 80/200 | Train loss: 0.133007 | Test Accuracy: 0.845
Epoch: 81/200 | Train loss: 0.130577 | Test Accuracy: 0.845
Epoch: 82/200 | Train loss: 0.128223 | Test Accuracy: 0.845
Epoch: 83/200 | Train loss: 0.125883 | Test Accuracy: 0.845
Epoch: 84/200 | Train loss: 0.123616 | Test Accuracy: 0.845
Epoch: 85/200 | Train loss: 0.121388 | Test Accuracy: 0.845
Epoch: 86/200 | Train loss: 0.119203 | Test Accuracy: 0.845
Epoch: 87/200 | Train loss: 0.117094 | Test Accuracy: 0.845
Epoch: 88/200 | Train loss: 0.115035 | Test Accuracy: 0.845
Epoch: 89/200 | Train loss: 0.113012 | Test Accuracy: 0.845
Epoch: 90/200 | Train loss: 0.111049 | Test Accuracy: 0.845
Epoch: 91/200 | Train loss: 0.109139 | Test Accuracy: 0.845
Epoch: 92/200 | Train loss: 0.107273 | Test Accuracy: 0.845
Epoch: 93/200 | Train loss: 0.105428 | Test Accuracy: 0.845
Epoch: 94/200 | Train loss: 0.103629 | Test Accuracy: 0.845
Epoch: 95/200 | Train loss: 0.101858 | Test Accuracy: 0.845
Epoch: 96/200 | Train loss: 0.100169 | Test Accuracy: 0.845
Epoch: 97/200 | Train loss: 0.0984903 | Test Accuracy: 0.845
Epoch: 98/200 | Train loss: 0.0968328 | Test Accuracy: 0.845
Epoch: 99/200 | Train loss: 0.0952505 | Test Accuracy: 0.845
Epoch: 100/200 | Train loss: 0.0936432 | Test Accuracy: 0.85
Epoch: 101/200 | Train loss: 0.0921455 | Test Accuracy: 0.85
Epoch: 102/200 | Train loss: 0.090617 | Test Accuracy: 0.85
Epoch: 103/200 | Train loss: 0.0891499 | Test Accuracy: 0.85
Epoch: 104/200 | Train loss: 0.0876947 | Test Accuracy: 0.85
Epoch: 105/200 | Train loss: 0.0862966 | Test Accuracy: 0.85
Epoch: 106/200 | Train loss: 0.0849275 | Test Accuracy: 0.85
Epoch: 107/200 | Train loss: 0.083552 | Test Accuracy: 0.85
```

```
Epoch: 108/200 | Train loss: 0.0822533 | Test Accuracy: 0.85
Epoch: 109/200 | Train loss: 0.0809346 | Test Accuracy: 0.85
Epoch: 110/200 | Train loss: 0.0796778 | Test Accuracy: 0.85
Epoch: 111/200 | Train loss: 0.0784252 | Test Accuracy: 0.85
Epoch: 112/200 | Train loss: 0.0771994 | Test Accuracy: 0.85
Epoch: 113/200 | Train loss: 0.0760304 | Test Accuracy: 0.85
Epoch: 114/200 | Train loss: 0.0748484 | Test Accuracy: 0.85
Epoch: 115/200 | Train loss: 0.0737013 | Test Accuracy: 0.85
Epoch: 116/200 | Train loss: 0.0725929 | Test Accuracy: 0.85
Epoch: 117/200 | Train loss: 0.0714964 | Test Accuracy: 0.85
Epoch: 118/200 | Train loss: 0.0704167 | Test Accuracy: 0.85
Epoch: 119/200 | Train loss: 0.0693614 | Test Accuracy: 0.85
Epoch: 120/200 | Train loss: 0.0683221 | Test Accuracy: 0.85
Epoch: 121/200 | Train loss: 0.0673195 | Test Accuracy: 0.85
Epoch: 122/200 | Train loss: 0.0663385 | Test Accuracy: 0.85
Epoch: 123/200 | Train loss: 0.0653571 | Test Accuracy: 0.85
Epoch: 124/200 | Train loss: 0.0643872 | Test Accuracy: 0.85
Epoch: 125/200 | Train loss: 0.0634374 | Test Accuracy: 0.85
Epoch: 126/200 | Train loss: 0.0625282 | Test Accuracy: 0.85
Epoch: 127/200 | Train loss: 0.0616318 | Test Accuracy: 0.85
Epoch: 128/200 | Train loss: 0.060734 | Test Accuracy: 0.85
Epoch: 129/200 | Train loss: 0.0598637 | Test Accuracy: 0.85
Epoch: 130/200 | Train loss: 0.0590068 | Test Accuracy: 0.85
Epoch: 131/200 | Train loss: 0.0581535 | Test Accuracy: 0.85
Epoch: 132/200 | Train loss: 0.057344 | Test Accuracy: 0.85
Epoch: 133/200 | Train loss: 0.0565305 | Test Accuracy: 0.85
Epoch: 134/200 | Train loss: 0.0557323 | Test Accuracy: 0.85
Epoch: 135/200 | Train loss: 0.0549596 | Test Accuracy: 0.85
Epoch: 136/200 | Train loss: 0.054195 | Test Accuracy: 0.85
Epoch: 137/200 | Train loss: 0.0534396 | Test Accuracy: 0.85
Epoch: 138/200 | Train loss: 0.0526935 | Test Accuracy: 0.85
Epoch: 139/200 | Train loss: 0.0519834 | Test Accuracy: 0.85
Epoch: 140/200 | Train loss: 0.0512721 | Test Accuracy: 0.85
Epoch: 141/200 | Train loss: 0.0505676 | Test Accuracy: 0.85
Epoch: 142/200 | Train loss: 0.0498803 | Test Accuracy: 0.85
Epoch: 143/200 | Train loss: 0.049214 | Test Accuracy: 0.85
Epoch: 144/200 | Train loss: 0.0485525 | Test Accuracy: 0.85
Epoch: 145/200 | Train loss: 0.0478794 | Test Accuracy: 0.85
Epoch: 146/200 | Train loss: 0.0472655 | Test Accuracy: 0.845
Epoch: 147/200 | Train loss: 0.0466187 | Test Accuracy: 0.85
Epoch: 148/200 | Train loss: 0.0460011 | Test Accuracy: 0.85
Epoch: 149/200 | Train loss: 0.0454059 | Test Accuracy: 0.85
Epoch: 150/200 | Train loss: 0.0447997 | Test Accuracy: 0.85
Epoch: 151/200 | Train loss: 0.0442234 | Test Accuracy: 0.85
Epoch: 152/200 | Train loss: 0.0436382 | Test Accuracy: 0.85
Epoch: 153/200 | Train loss: 0.0430743 | Test Accuracy: 0.85
Epoch: 154/200 | Train loss: 0.0425156 | Test Accuracy: 0.845
Epoch: 155/200 | Train loss: 0.04198 | Test Accuracy: 0.85
```

```
Epoch: 156/200 | Train loss: 0.0414365 | Test Accuracy: 0.845
Epoch: 157/200 | Train loss: 0.0409135 | Test Accuracy: 0.845
Epoch: 158/200 | Train loss: 0.0403928 | Test Accuracy: 0.85
Epoch: 159/200 | Train loss: 0.039884 | Test Accuracy: 0.85
Epoch: 160/200 | Train loss: 0.0393841 | Test Accuracy: 0.845
Epoch: 161/200 | Train loss: 0.0388807 | Test Accuracy: 0.85
Epoch: 162/200 | Train loss: 0.038407 | Test Accuracy: 0.845
Epoch: 163/200 | Train loss: 0.0379119 | Test Accuracy: 0.85
Epoch: 164/200 | Train loss: 0.0374498 | Test Accuracy: 0.85
Epoch: 165/200 | Train loss: 0.0369856 | Test Accuracy: 0.85
Epoch: 166/200 | Train loss: 0.036529 | Test Accuracy: 0.85
Epoch: 167/200 | Train loss: 0.0360794 | Test Accuracy: 0.85
Epoch: 168/200 | Train loss: 0.0356333 | Test Accuracy: 0.85
Epoch: 169/200 | Train loss: 0.0351992 | Test Accuracy: 0.85
Epoch: 170/200 | Train loss: 0.0347689 | Test Accuracy: 0.85
Epoch: 171/200 | Train loss: 0.0343529 | Test Accuracy: 0.85
Epoch: 172/200 | Train loss: 0.0339328 | Test Accuracy: 0.845
Epoch: 173/200 | Train loss: 0.033529 | Test Accuracy: 0.85
Epoch: 174/200 | Train loss: 0.0331239 | Test Accuracy: 0.85
Epoch: 175/200 | Train loss: 0.0327245 | Test Accuracy: 0.85
Epoch: 176/200 | Train loss: 0.0323352 | Test Accuracy: 0.85
Epoch: 177/200 | Train loss: 0.0319534 | Test Accuracy: 0.85
Epoch: 178/200 | Train loss: 0.0315785 | Test Accuracy: 0.85
Epoch: 179/200 | Train loss: 0.0312105 | Test Accuracy: 0.85
Epoch: 180/200 | Train loss: 0.0308438 | Test Accuracy: 0.85
Epoch: 181/200 | Train loss: 0.0304813 | Test Accuracy: 0.85
Epoch: 182/200 | Train loss: 0.0301249 | Test Accuracy: 0.85
Epoch: 183/200 | Train loss: 0.029781 | Test Accuracy: 0.85
Epoch: 184/200 | Train loss: 0.0294336 | Test Accuracy: 0.85
Epoch: 185/200 | Train loss: 0.0290973 | Test Accuracy: 0.85
Epoch: 186/200 | Train loss: 0.0287597 | Test Accuracy: 0.85
Epoch: 187/200 | Train loss: 0.0284328 | Test Accuracy: 0.85
Epoch: 188/200 | Train loss: 0.0281109 | Test Accuracy: 0.85
Epoch: 189/200 | Train loss: 0.0277851 | Test Accuracy: 0.85
Epoch: 190/200 | Train loss: 0.0274793 | Test Accuracy: 0.85
Epoch: 191/200 | Train loss: 0.0271657 | Test Accuracy: 0.85
Epoch: 192/200 | Train loss: 0.0268628 | Test Accuracy: 0.85
Epoch: 193/200 | Train loss: 0.0265607 | Test Accuracy: 0.85
Epoch: 194/200 | Train loss: 0.0262676 | Test Accuracy: 0.85
Epoch: 195/200 | Train loss: 0.0259738 | Test Accuracy: 0.85
Epoch: 196/200 | Train loss: 0.025689 | Test Accuracy: 0.85
Epoch: 197/200 | Train loss: 0.0253992 | Test Accuracy: 0.85
Epoch: 198/200 | Train loss: 0.0251232 | Test Accuracy: 0.85
Epoch: 199/200 | Train loss: 0.0248432 | Test Accuracy: 0.85
Epoch: 200/200 | Train loss: 0.0245696 | Test Accuracy: 0.85
```